



VMware vSphere Infrastructure Deployment

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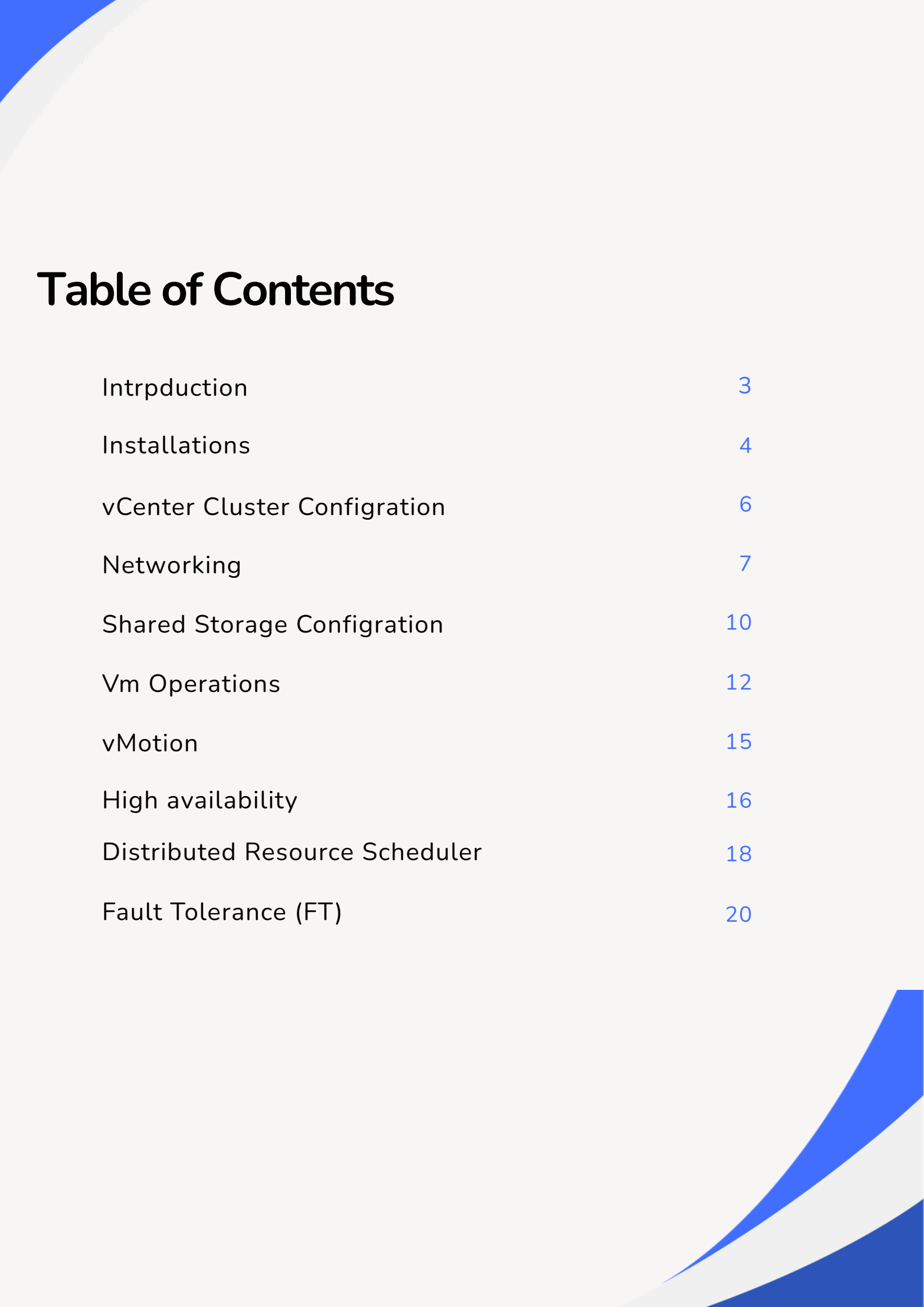


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1. Introduction

- This project focuses on building a complete vSphere environment using VMware Workstation as a lab platform. The environment includes multiple ESXi hosts connected to a centralized vCenter Server, allowing the implementation of advanced VMware features such as vMotion, High Availability (HA), Distributed Resource Scheduler (DRS), Fault Tolerance (FT), and shared storage management.
- The main goal of this project is to simulate a real enterprise virtualization infrastructure and gain practical experience in deploying, configuring, managing, and testing VMware vSphere technologies. Throughout this project, virtual machines were created and managed, networking and storage were configured, and cluster services were tested to ensure availability and workload balancing across hosts.

1.1 Project Overview

This project demonstrates the deployment and configuration of a VMware vSphere virtualized infrastructure in a lab environment. Two ESXi hosts were installed as virtual machines inside VMware Workstation and managed centrally through VMware vCenter Server Appliance (VCSA).

The ESXi hosts were combined into a cluster to enable enterprise features including:

- vMotion for live virtual machine migration
- High Availability (HA) for automatic VM recovery during host failure
- Distributed Resource Scheduler (DRS) for workload balancing
- Fault Tolerance (FT) for continuous VM availability

In addition, the project included the configuration of virtual networking using multiple vSwitches and port groups, setup of NFS shared storage, creation of a content library for ISO management, and deployment of virtual machines running Windows or Linux operating systems.

Several administrative operations were performed such as VM cloning, snapshot creation, template management, and VM migration between hosts. Testing scenarios were also implemented to verify the functionality of HA, DRS, vMotion, and FT within the cluster environment.

The project provides hands-on experience with VMware virtualization technologies and demonstrates how enterprise virtual infrastructures are designed, configured, and maintained.

2. Installtions

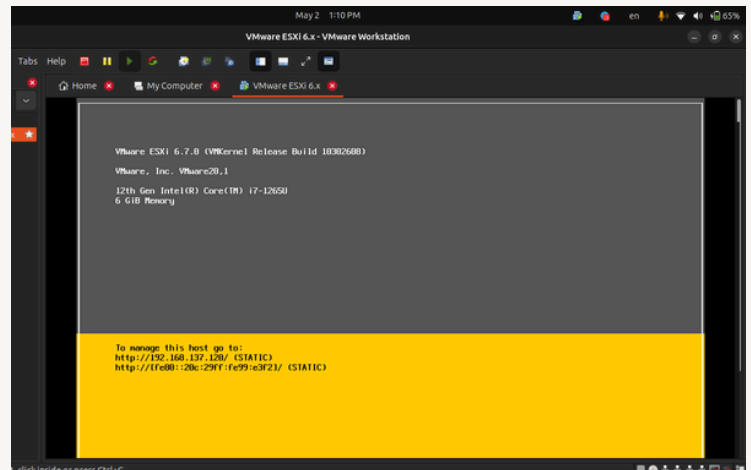
1-ESXi Hosts Installation

In this phase of the project, the ESXi hosts were installed as virtual machines within VMware Workstation to create the virtualization infrastructure required for the vSphere environment. Each ESXi host was configured with the necessary CPU, memory, storage, and networking resources to support cluster operations and virtual machine management. After the installation process was completed, the management network settings such as IP address, subnet mask, gateway, and DNS configuration were assigned to allow communication between the ESXi hosts and vCenter Server.

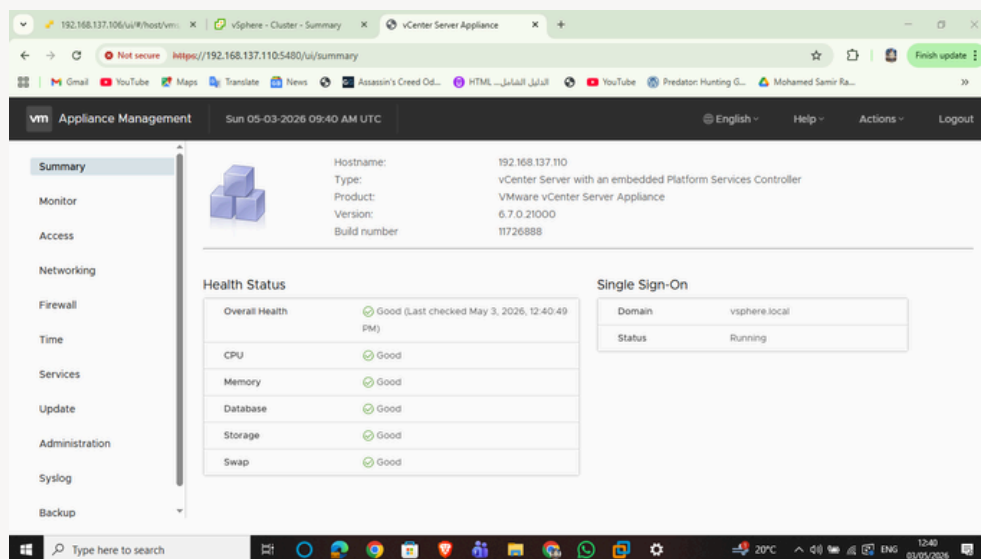
2-vCenter Server Deployment on ESXi-0

After preparing the ESXi hosts, the VMware vCenter Server Appliance (VCSA) was deployed on ESXi-0 to provide centralized management for the virtual infrastructure. The deployment process included configuring the appliance settings, assigning network parameters, and completing the initial setup wizard. Once deployment was finished, vCenter Server was used to manage the ESXi hosts, create clusters, configure advanced vSphere features, and monitor the overall environment.

- 2 esxi hosts are created



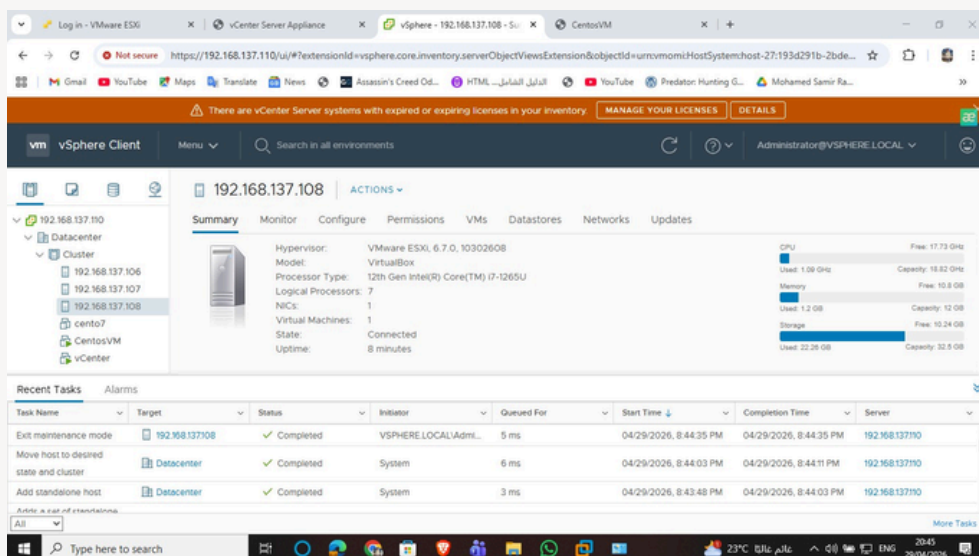
- vCenter



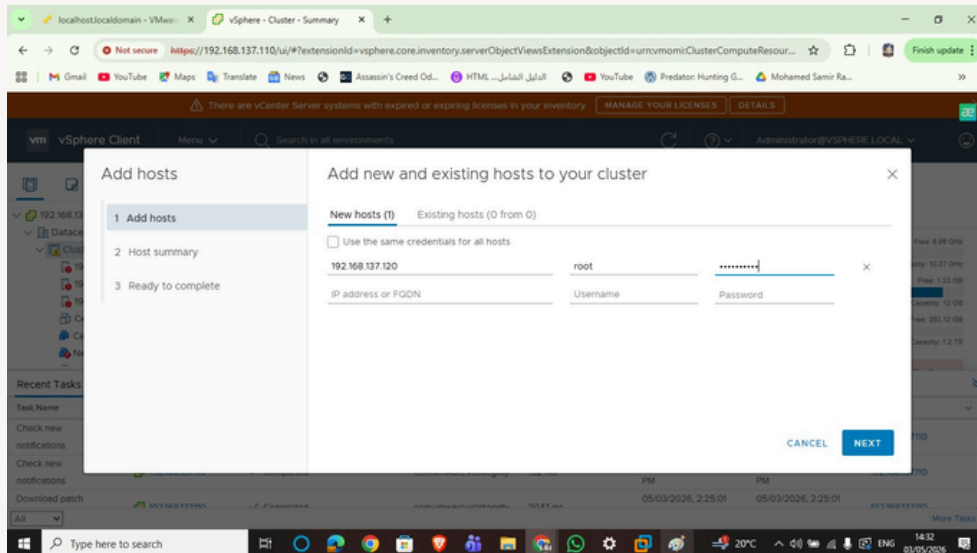
4. vCenter Cluster Configuration

- After successfully deploying and connecting the ESXi hosts to vCenter Server, a cluster was created to enable centralized management and advanced vSphere features. The cluster allows multiple ESXi hosts to work together as a single resource pool, improving availability, performance, and workload distribution within the virtual environment.
- During this configuration, both ESXi hosts were added to the cluster through vCenter Server. Advanced services such as vMotion, High Availability (HA), and Distributed Resource Scheduler (DRS) were enabled and configured to support live virtual machine migration, automatic recovery during host failure, and intelligent workload balancing across hosts.
- The cluster configuration also included verifying network connectivity, shared storage accessibility, and compatibility settings to ensure proper operation of VMware enterprise features. This setup provides a more reliable and efficient virtualization infrastructure similar to production enterprise environments.

- cluster configuration



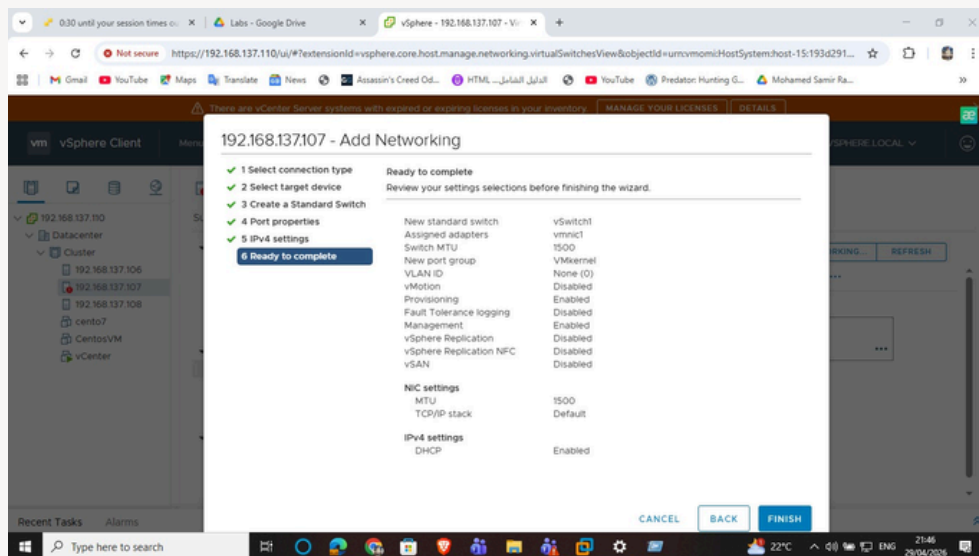
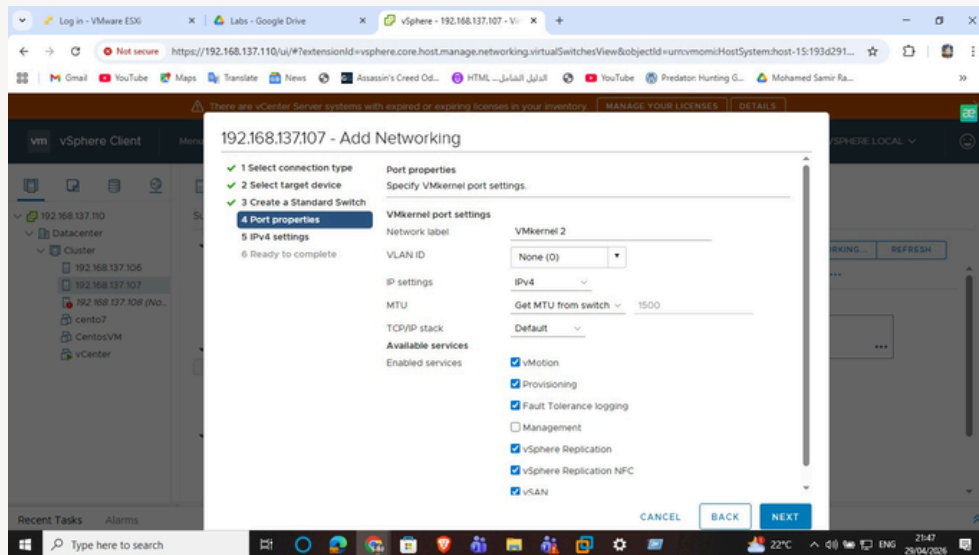
- adding host to cluster



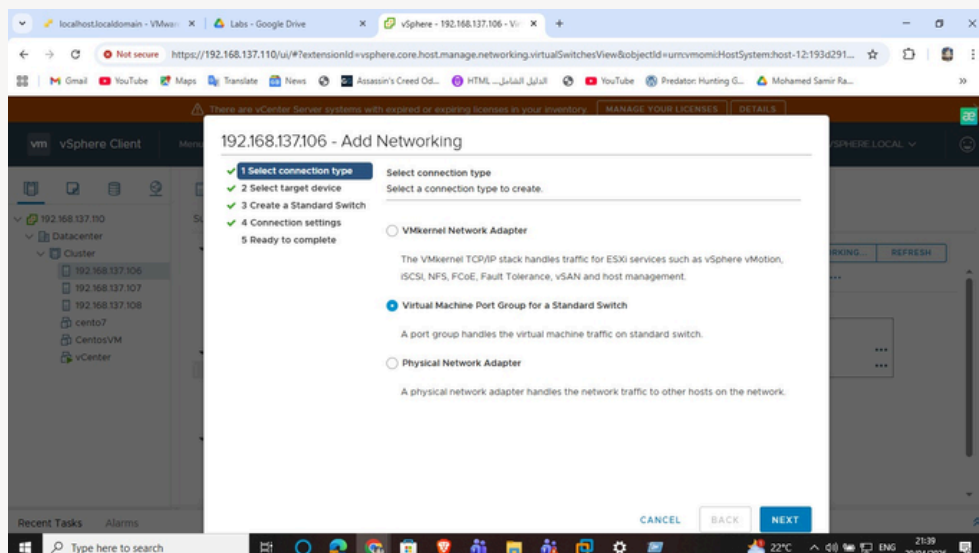
4. Networking

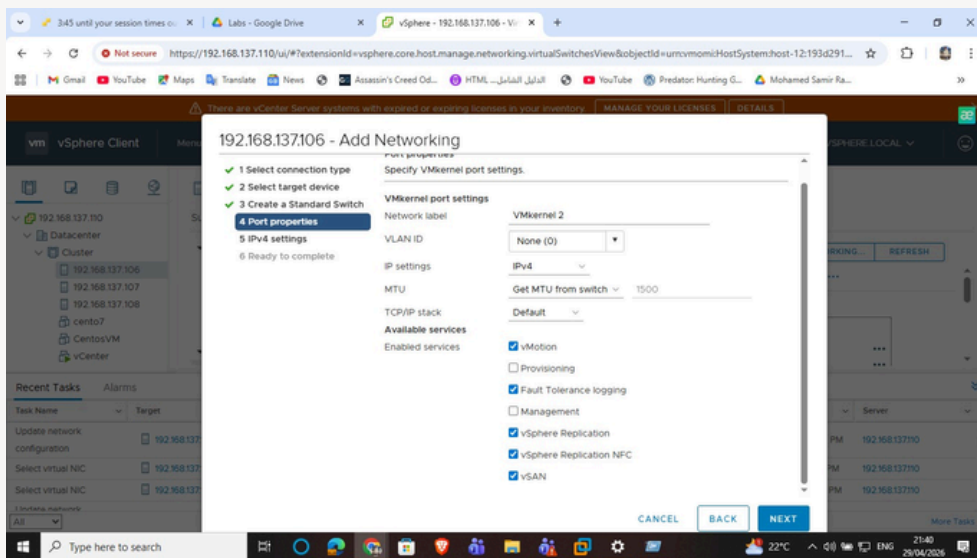
- In this section, virtual networking was configured to provide communication between the ESXi hosts, virtual machines, storage, and management services within the vSphere environment. VMware Standard vSwitches were created and configured on each ESXi host to separate different types of network traffic and improve network organization and performance.
- A dedicated vSwitch was configured for management and virtual machine traffic, including port groups used for ESXi management access and VM network communication. Another vSwitch was created specifically for storage and vMotion traffic to isolate migration and storage operations from the main management network.
- Port groups were assigned with appropriate network settings and virtual adapters to ensure proper connectivity between hosts and services. This network design helps improve security, performance, and stability within the virtual infrastructure while supporting advanced VMware features such as vMotion, HA, and FT.

- vmkernel

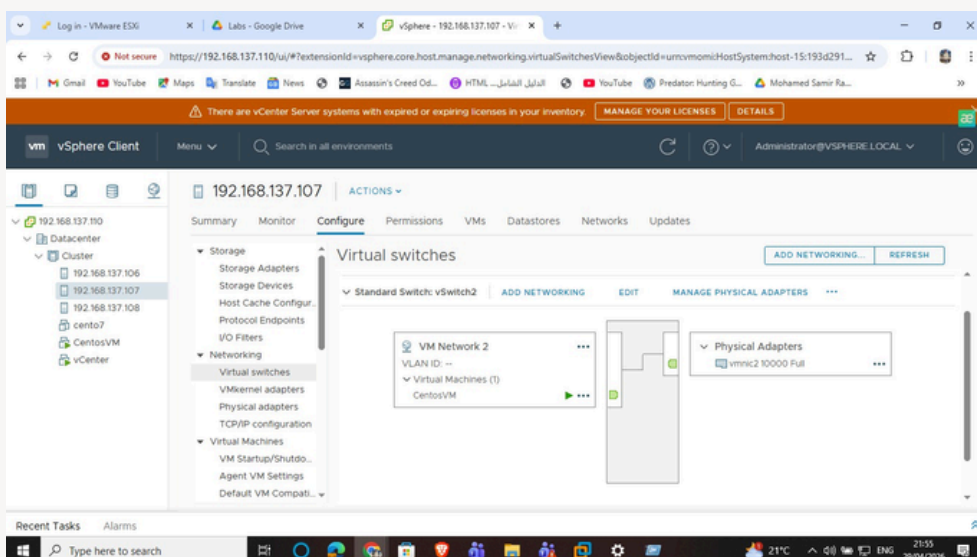
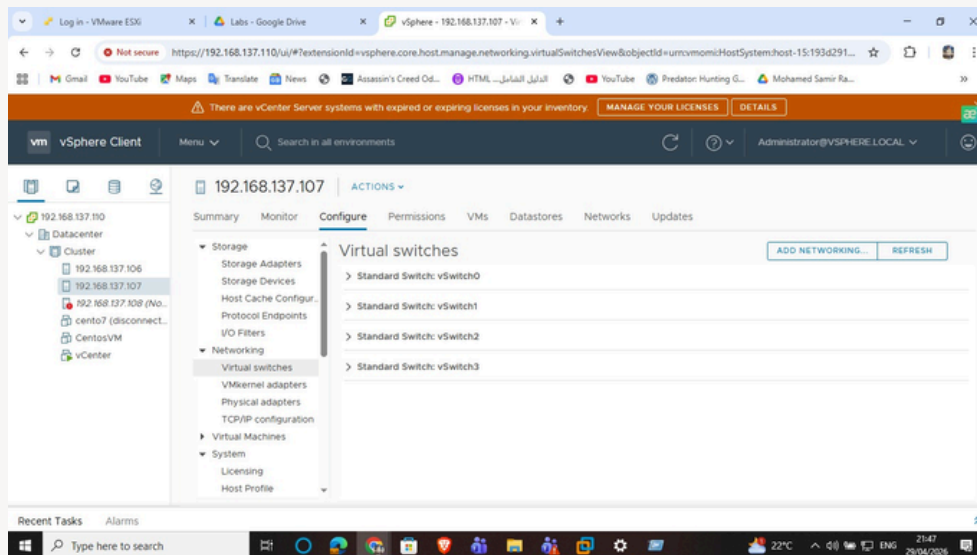


- port group





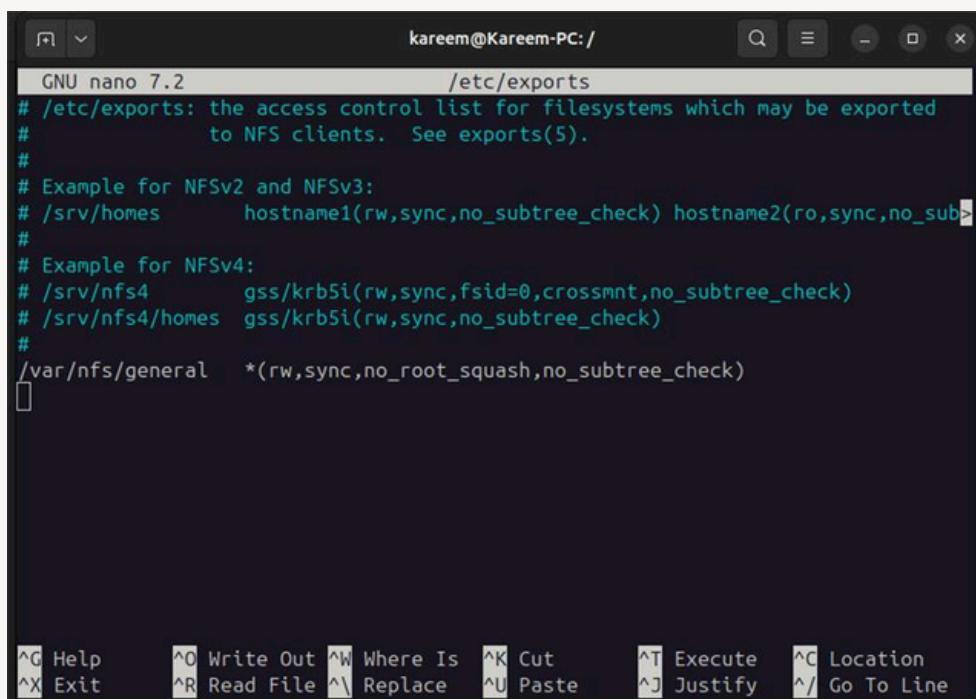
- switches are created



5. Shared Storage Configuration

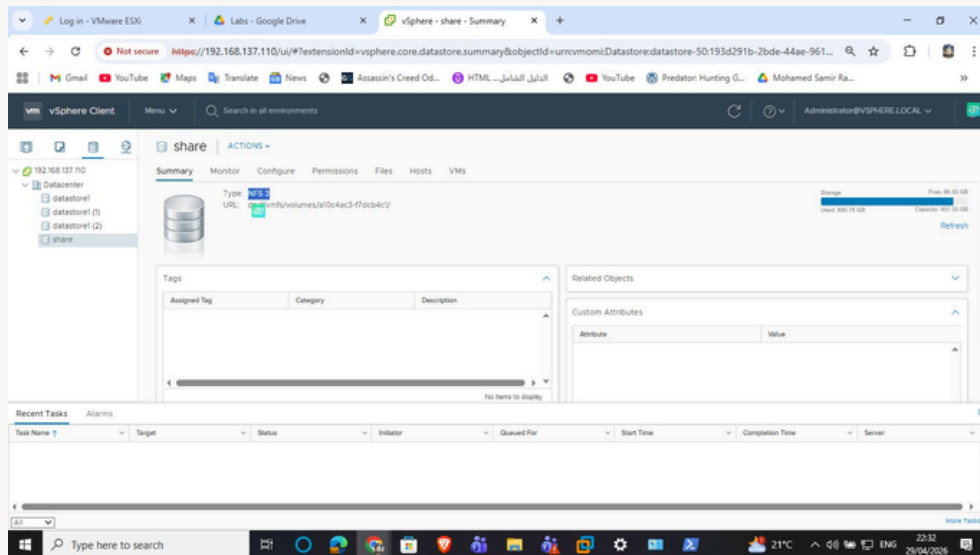
- Shared storage was configured in the vSphere environment using an NFS datastore to allow both ESXi hosts to access the same storage resources. Shared storage is required for advanced vSphere features such as vMotion, High Availability (HA), Fault Tolerance (FT), and Distributed Resource Scheduler (DRS).
- The NFS storage server was configured and connected to both ESXi hosts through the storage network. In vCenter, a new NFS datastore was created by specifying the server IP address, shared folder path, and datastore name.
- After the configuration was completed, the datastore became accessible from both ESXi hosts and was used to store virtual machine files and ISO images. Connectivity and accessibility were verified to ensure that all hosts could use the shared storage successfully.

- NFS server export configuration

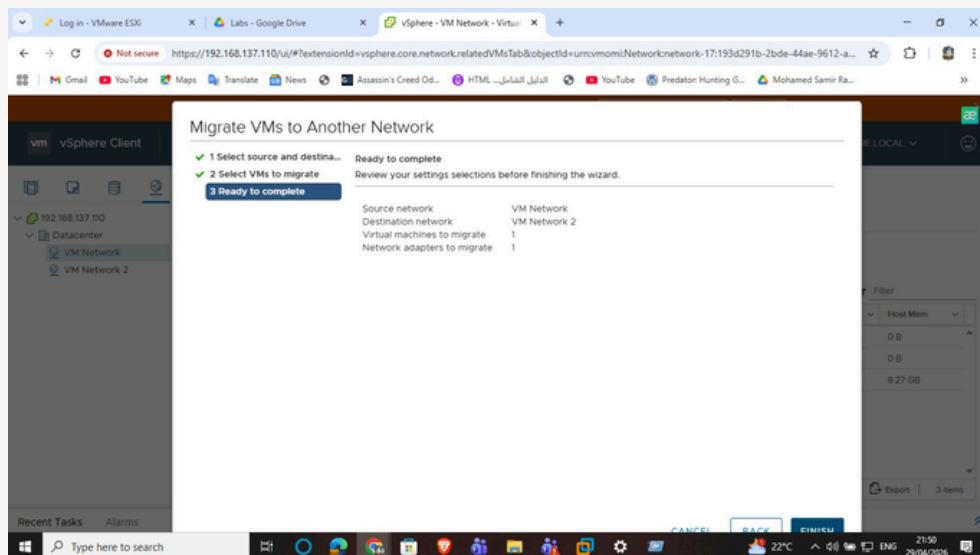


```
kareem@Kareem-PC: /  
GNU nano 7.2 /etc/exports  
# /etc/exports: the access control list for filesystems which may be exported  
# to NFS clients.  See exports(5).  
#  
# Example for NFSv2 and NFSv3:  
# /srv/homes      hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_subtree_check)  
#  
# Example for NFSv4:  
# /srv/nfs4       gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)  
# /srv/nfs4/homes gss/krb5i(rw,sync,no_subtree_check)  
#  
/var/nfs/general *(rw,sync,no_root_squash,no_subtree_check)  
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

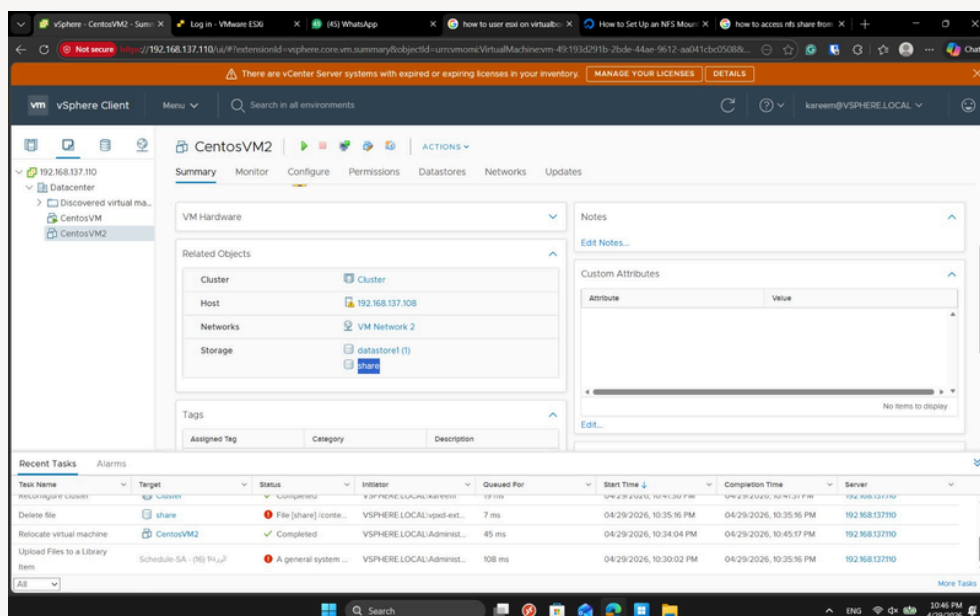
- NFS Datastore Added to vSphere



- migrate vms to another network



- CentOS NFS storage server



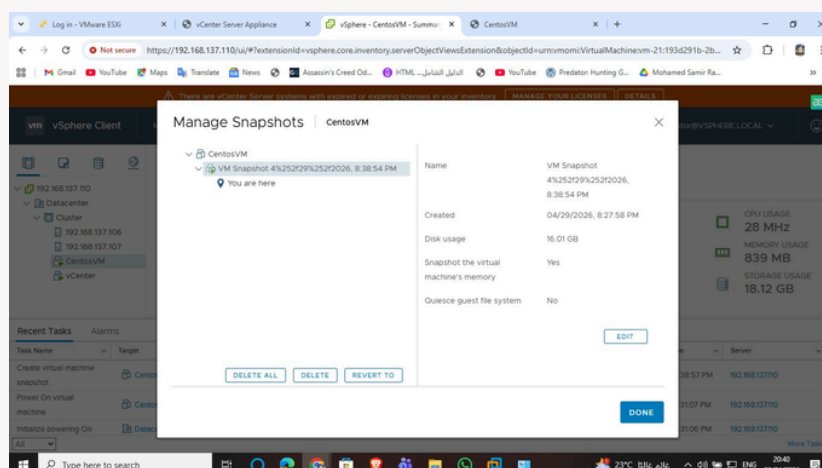
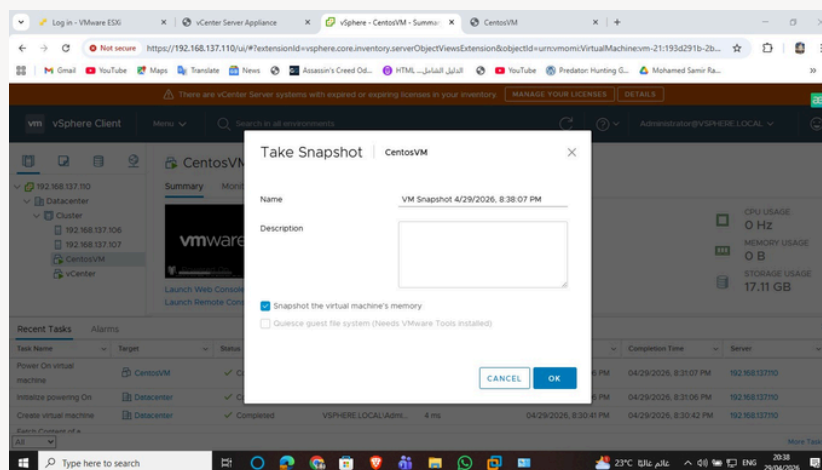
6. VM Operations

6.1 Snapshot

One of the main operations performed on virtual machines is creating snapshots. A snapshot saves the current state of a VM at a specific time, including its disk, memory, and settings.

In vCenter, the deployed VM was selected and the Take Snapshot option was used before making any changes. This is useful for testing and troubleshooting because the VM can be restored to its previous state if something goes wrong. After creating the snapshot, it was checked in the Snapshot Manager to confirm that it was created successfully and ready for rollback if needed.

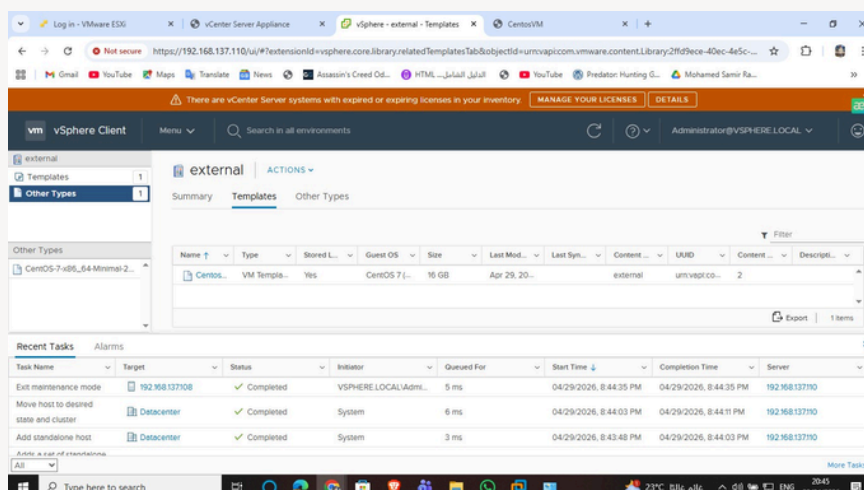
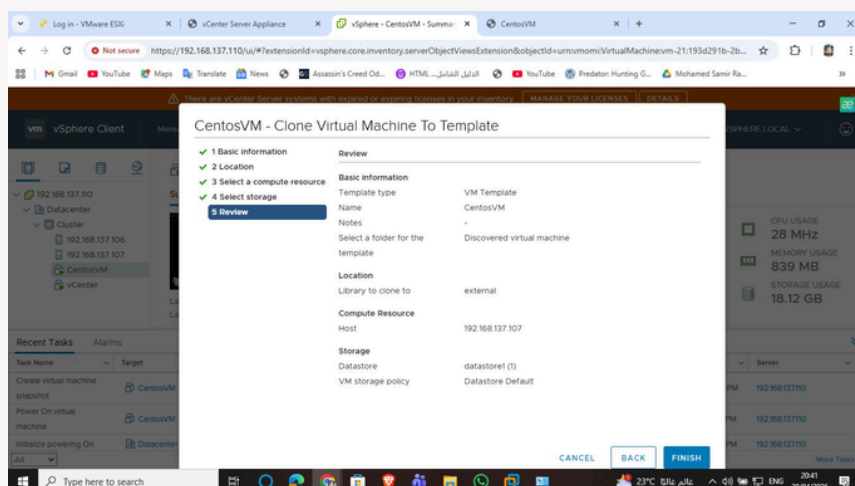
- taking and managing snapshot



6.2 Clone to template

Another important operation is converting a virtual machine into a template. In vCenter, the deployed VM was selected and the Clone to Template option was used to create a reusable image of the VM.

During this process, a customization specification was applied to set basic configuration details like the computer name and network settings. This helps when deploying new VMs so they can be configured automatically without manual setup. After completion, the template appeared in the vCenter inventory and was ready to be used to deploy multiple identical virtual machines quickly.



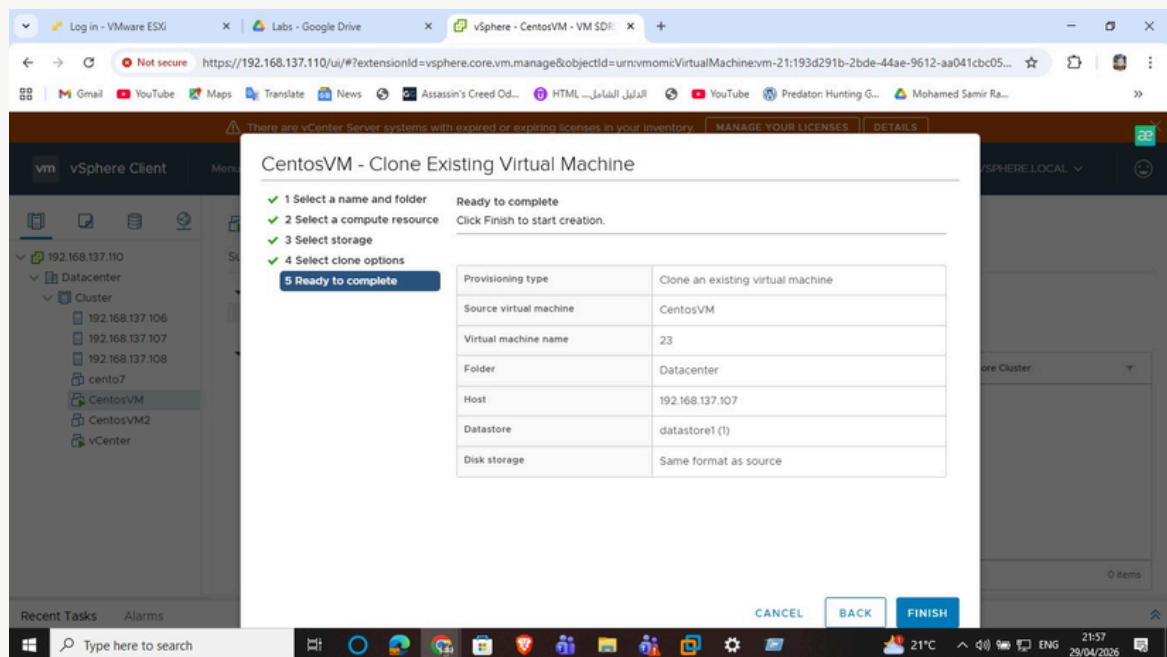
6.3 Clone

Cloning a virtual machine creates an exact copy of an existing VM, including its operating system and configuration.

In vCenter, the deployed VM was selected and the Clone option was used. The destination ESXi host and datastore were chosen to place the new VM inside the cluster environment.

After the process finished, the cloned VM appeared in the inventory and worked as a separate independent machine from the original VM.

- cloning existing virtual machine



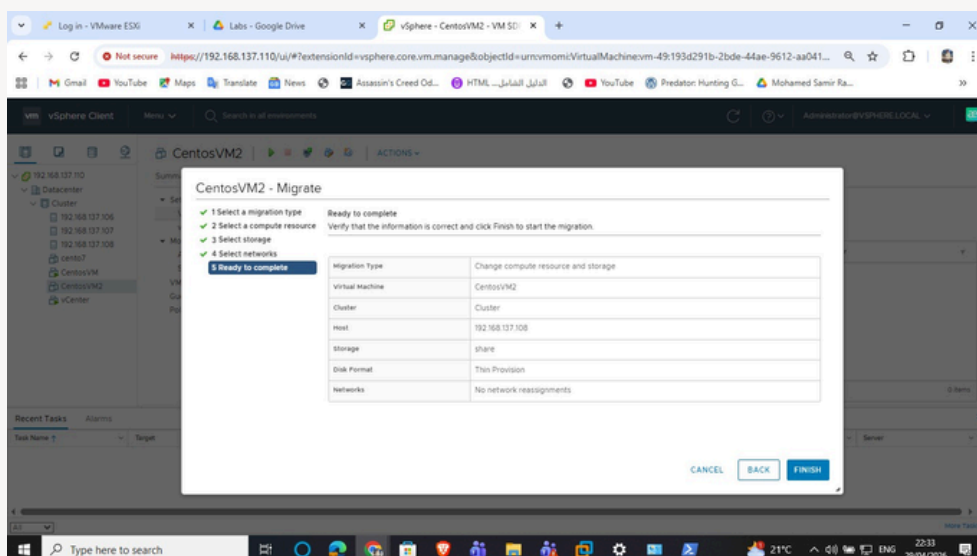
7. vMotion

Cloning a virtual machine creates an exact copy of an existing VM, including its operating system and configuration.

In vCenter, the deployed VM was selected and the Clone option was used. The destination ESXi host and datastore were chosen to place the new VM inside the cluster environment.

After the process finished, the cloned VM appeared in the inventory and worked as a separate independent machine from the original VM.

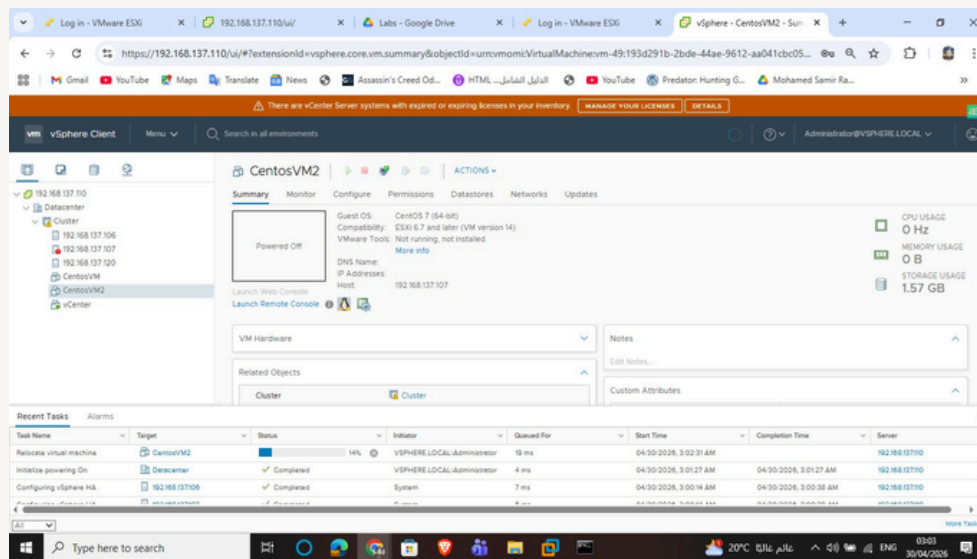
- Configuring VM Migration Settings



- Migration Resource Selection

Task Name	Target	Status	Initiator	Queued For	Start Time	Completion Time	Server
Delete file	share	File [share] /conte...	VSPHERE.LOCAL/vsped-ext...	7 ms	04/29/2026, 10:35:16 PM	04/29/2026, 10:35:16 PM	192.168.137.100
Relocate virtual machine	CentosVM2	45%	VSPHERE.LOCAL/Administ...	45 ms	04/29/2026, 10:34:04 PM		192.168.137.100
Upload Files to a Library Item	Schedule SA - (R) Ph...	A general system ...	VSPHERE.LOCAL/Administ...	108 ms	04/29/2026, 10:30:02 PM	04/29/2026, 10:35:16 PM	192.168.137.100

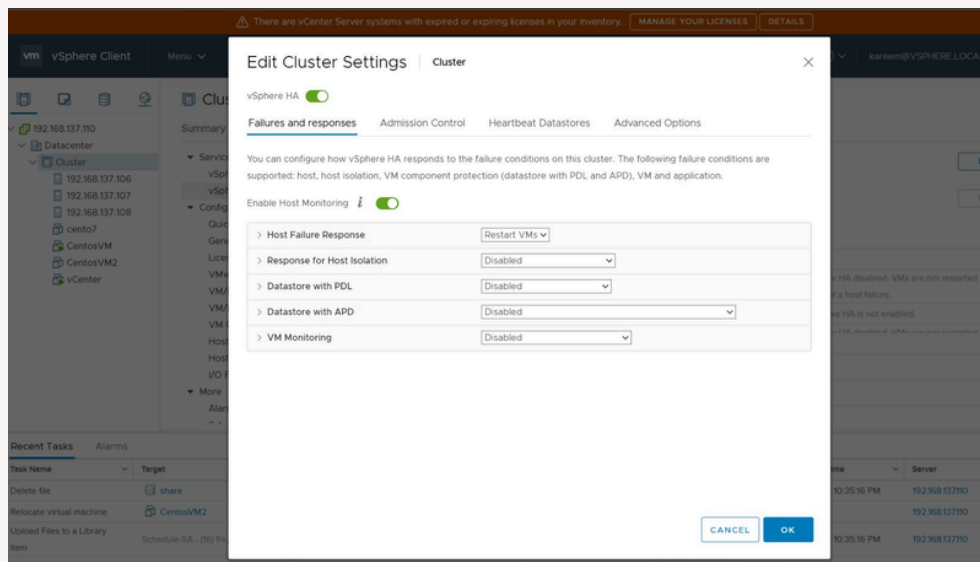
- VM Migration Completed Successfully



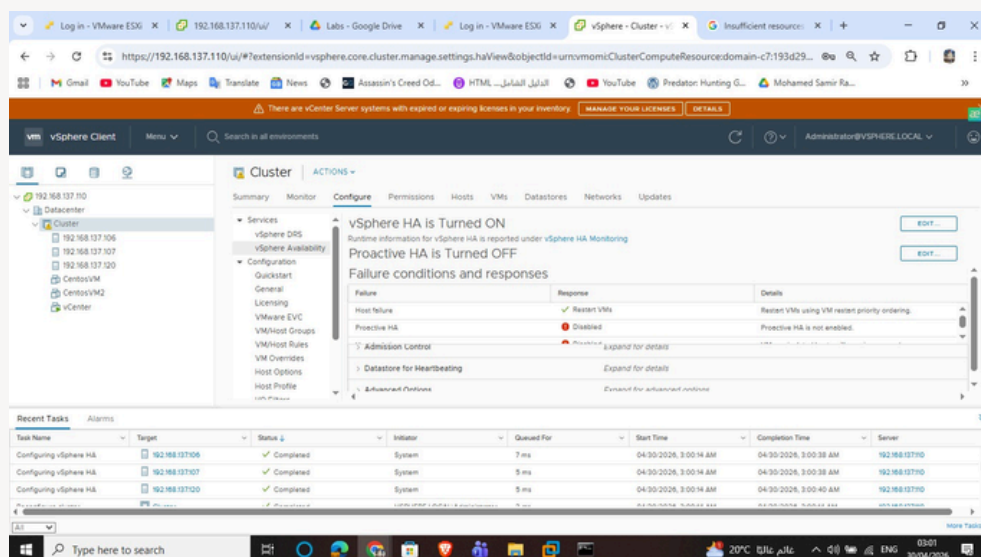
8. High Availability

- High Availability (HA) was configured in the vSphere cluster to provide automatic recovery of virtual machines in case of ESXi host failure. This feature helps improve system availability and minimize downtime within the virtual environment.
- HA was enabled from the cluster settings in vCenter Server after adding both ESXi hosts to the cluster. Shared storage and network connectivity were verified to ensure proper communication between the hosts.
- To test HA functionality, one of the ESXi hosts was intentionally shut down to simulate a failure scenario. After the failure was detected, the affected virtual machines were automatically restarted on the second ESXi host, confirming that HA was functioning successfully and providing failover protection for the environment.

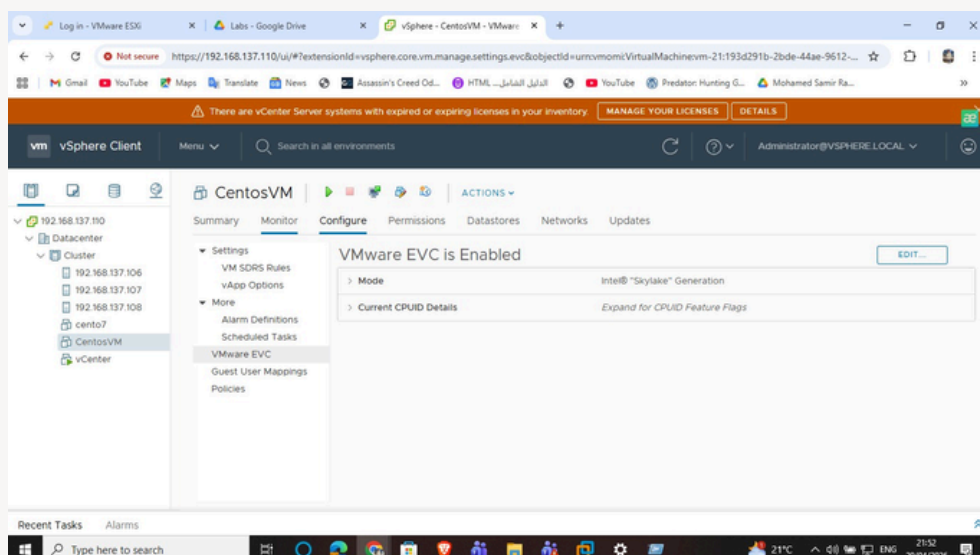
- editing virtual machine settings for HA support



- enabling availability features on the cluster



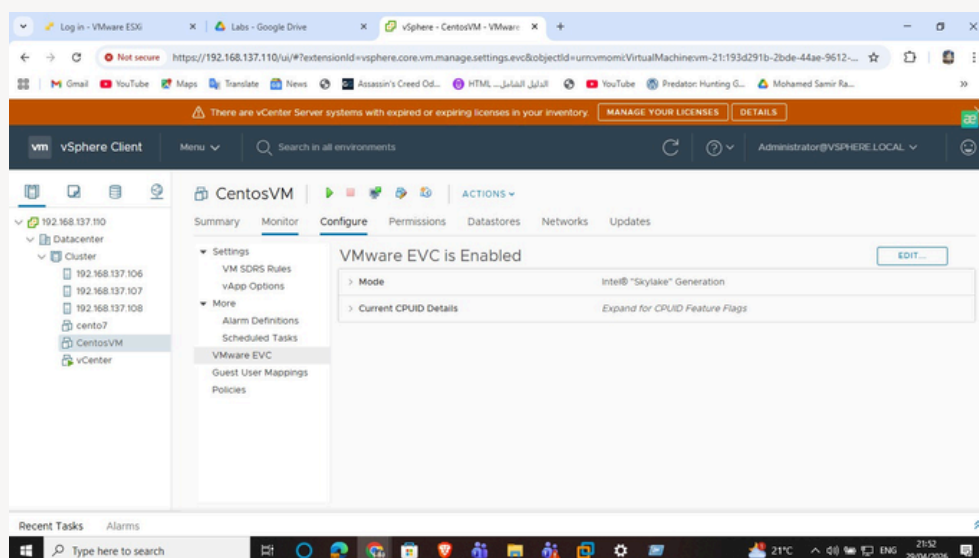
- VMware EVC configuration for cluster compatibility



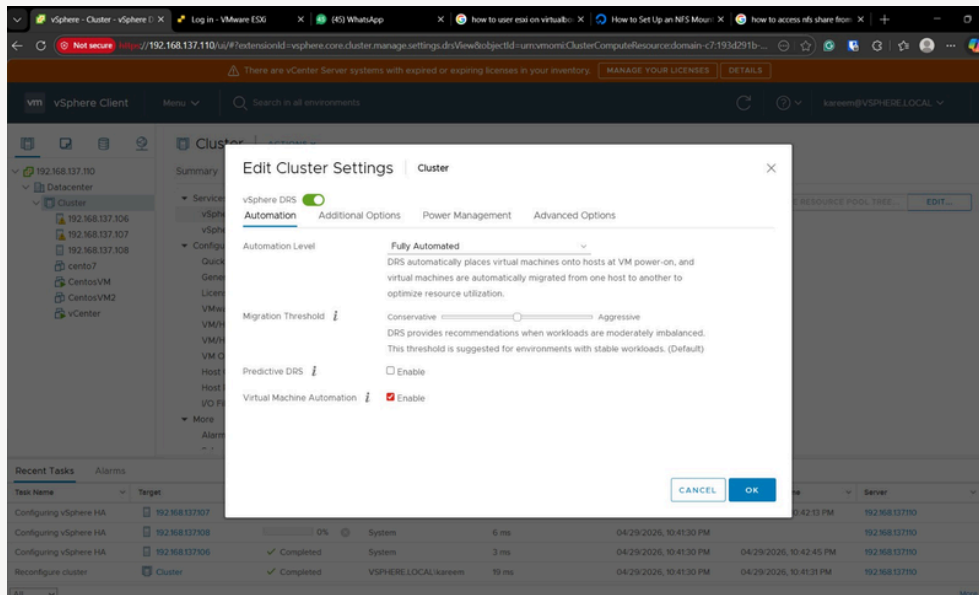
9. Distributed Resource Scheduler

- Distributed Resource Scheduler (DRS) was enabled in the vSphere cluster to automatically balance workloads and resource usage between the ESXi hosts. This feature helps improve performance by distributing virtual machines based on the available CPU and memory resources.
- DRS was configured from the cluster settings in vCenter Server after both ESXi hosts were added to the cluster. The automation level was set to automatically recommend and perform VM migrations when resource imbalance occurs.
- After enabling DRS, the cluster continuously monitored resource usage across the hosts. When one host became heavily utilized, DRS used vMotion to migrate virtual machines to the other host, ensuring better load balancing and efficient resource management within the environment.

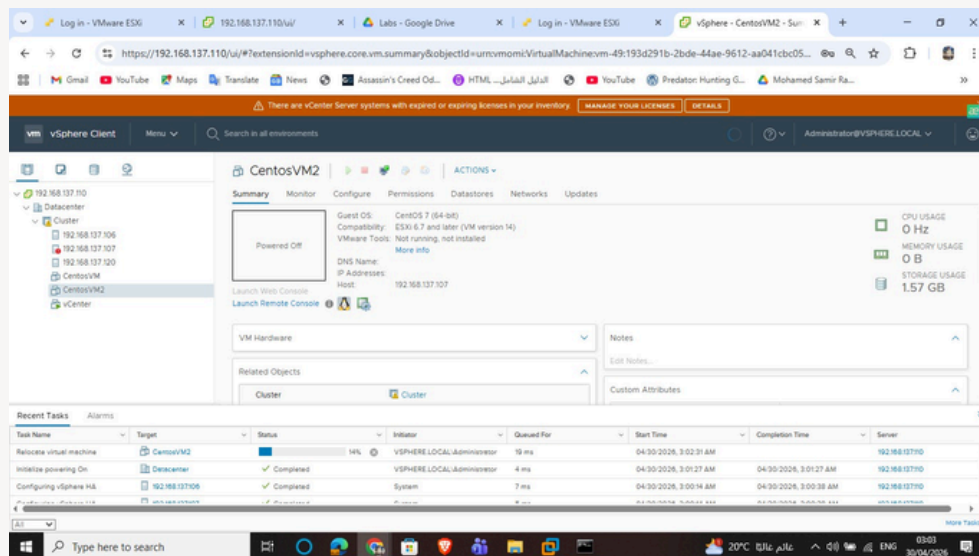
- EVC settings for DRS and vMotion support



- edit cluster setting to enable DRS



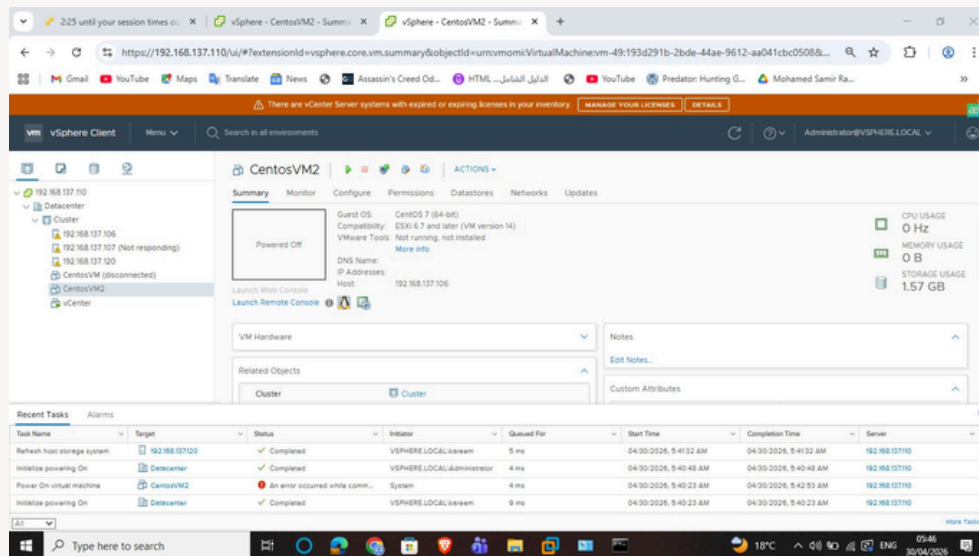
- VM2 on esxi 192.168.137.107



- turning off esxi



- VM2 had successfully migrated to esxi 192.168.137.106

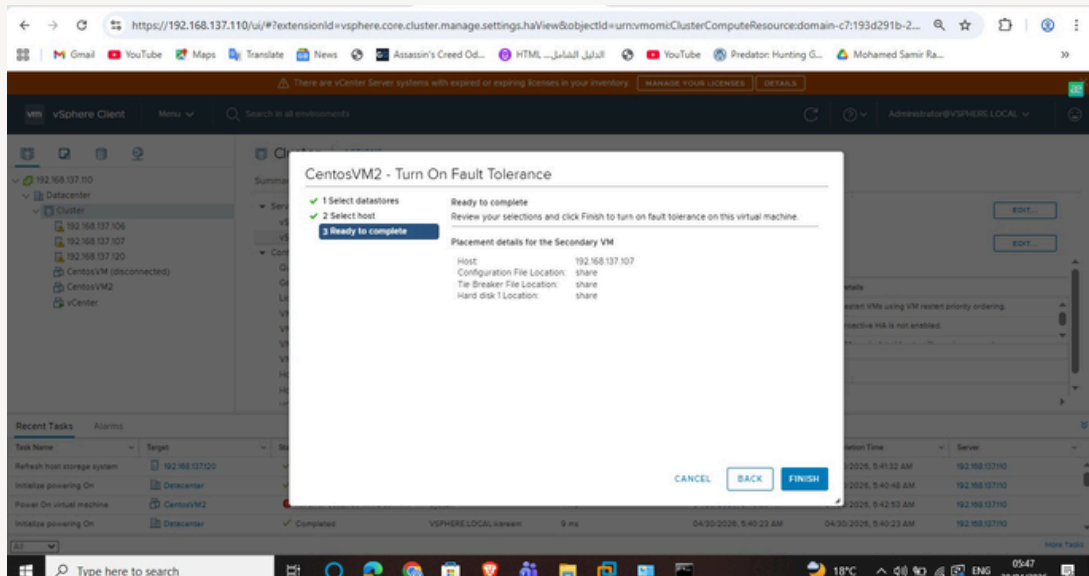


10. Fault Tolerance

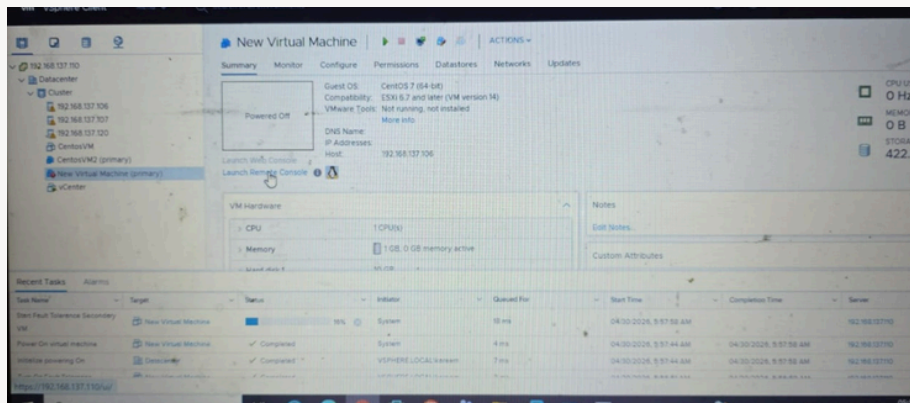
Fault Tolerance (FT) was configured in the vSphere environment to provide continuous availability for a virtual machine in case of host failure. FT works by creating a secondary copy of the VM on another ESXi host and keeping both virtual machines synchronized in real time.

Before enabling FT, the required configurations such as shared storage, compatible networking, and vMotion support were verified. In vCenter, the selected virtual machine was configured using the Turn On Fault Tolerance option. After enabling FT, the primary and secondary virtual machines were distributed across different ESXi hosts inside the cluster. To test the feature, one ESXi host was stopped to simulate a failure. The secondary VM immediately continued operation without interruption, confirming that Fault Tolerance was working successfully.

- activating fault tolerance on the VM



- Test fault tolerance





Thank you

